ALICE L WALTON SCHOOL OF MEDICINE

Transfusion Medicine

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Adapted from: Blood Transfusion

Learning Objectives (5) Versions

After completing this brick, you will be able to:

1. Recognize the antigens of the ABO and Rh blood groups.
2. Explain the clinical importance of antibodies involved in the ABO and Rh systems.
3. Describe blood typing, antibody screening, and cross-matching.
4. Distinguish commonly used blood products and their indications.
5. Differentiate potential complications of blood transfusion.

CASE CONNECTION

Toward the beginning of receiving a transfusion of packed red cells, HH, a 46-year-old woman in the medical ICU, develops chills and a temperature of 102.4°F. The nurse attributes these findings to her recent hospital-acquired pneumonia. Shortly thereafter, he notices HH is short of breath and hypotensive; you are called for an evaluation. Her blood pressure is 76/52 mm Hg, and you hear diffuse wheezing on pulmonary examination.

What is the likely diagnosis? How will you manage this condition? Consider your answers as you read, and we'll revisit at the end of the brick.

GO TO CONCLUSION ↓

What Is a Blood Type (LO #1)?

To explain what a blood type is, we need to begin by talking about antigens. Red blood cells (RBCs) have a lot of antigens on their surfaces (remember, an antigen is any substance that can trigger an immune response). Antigens that are similar to each other are put into the same group or system (these two words are used interchangeably). So a blood group is just a little collection of similar RBC antigens. There are a lot of different blood groups, but the two most clinically important are the ABO blood group and the Rh blood group.

A blood "type" refers to the combination of blood group antigens present on a person's RBCs. Typically, we only include antigens from the ABO and Rh group when referring to someone's blood type.

ABO System

There are three antigens involved in the ABO system (or group): A, B, and... no, not O. It's H. The H antigen is the precursor antigen present on the surface of all RBCs. Based on inherited alleles, enzymes may add specific sugars to the H antigen, resulting in the formation of A and B antigens on the RBC surface. Red blood cells that have only the A antigen on their surface are called type A red blood cells, and red blood cells that have only the B antigen are called type B. Red blood cells with both A and B antigens are called type AB, and if neither A nor B antigens are present, the red cells are called type O. Note that type O red cells still have the H antigen (even though we don't mention it in the blood type) (Figure 1).

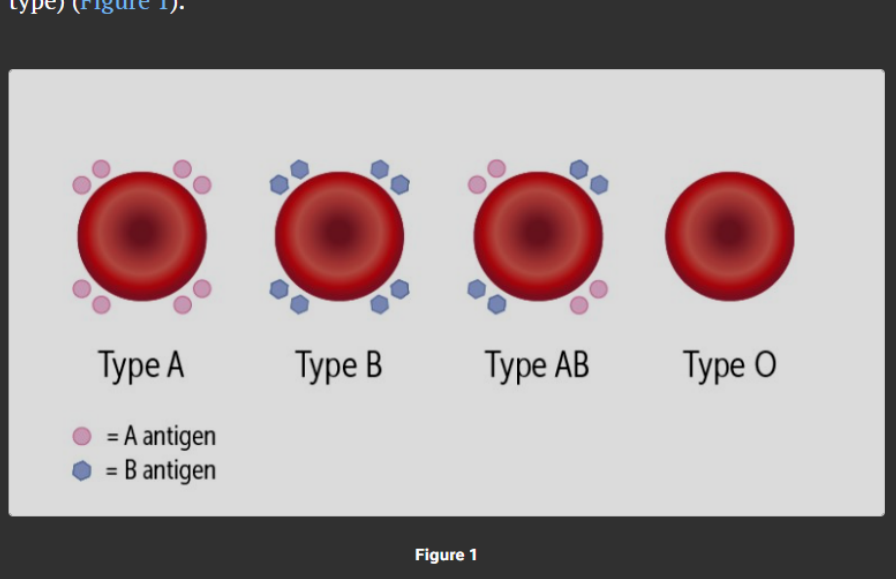

Figure 1

We inherit one allele (either A, B, or O) from each parent, and we express both alleles. This allows for six possible genotypes: AA, AO, BB, BO, AB, OO (Table 1). Each genotype leads to the expression of A antigen, B antigen, both A and B antigens, or neither antigen on the cell surface, leading to four antigen types and four blood group types.

Genotype	Antigen
AA	A antigen
AO	A antigen
BB	B antigen
BO	B antigen
AB	A and B antigens
OO	Neither A nor B antigen

Table 1

Rh System

This system contains three antigens, but the only one that is clinically important for us is the D antigen, which is also called the Rh antigen because it was first described in **Rh**esus monkeys (Rh means Rhesus factor). When red blood cells have the D antigen on their surface, they are said to be Rh-positive; if the D antigen is not present, the red blood cells are Rh-negative.

The antigens of the A [illegible] or absence (-) of the Rh antigen from the Rh group, can be combined into eight possible blood types: A-positive, A-negative, B-positive, B-negative, O-positive, O-negative, AB-positive, and AB-negative.

Antibodies (LO #2)

It's also important to understand that our immune system can form antibodies against the antigens in both the ABO and Rh groups (Table 2). Antibodies against the A and B antigens are naturally occurring, meaning that they are made automatically, without antigen exposure. This means we automatically make antibodies to whichever antigens (A or B) that our own red blood cells lack (for example, people with type A RBCs automatically have anti-B antibodies). Antibodies against the Rh (or D) antigen are not naturally occurring, which means that you only make anti-Rh antibodies if you lack the Rh antigen on your red blood cells and are exposed to Rh-positive blood (after transfusion or pregnancy with an Rh-positive fetus). If you have an antibody against one of these antigens, and you are given red blood cells that have that antigen on their surface, your antibodies will bind to the antigens on the transfused red blood cells, causing a potentially severe immunologic transfusion reaction (see below).

Table 2

Blood Type	Antigens on Red Blood Cells	Antibodies in Plasma
A-negative	A antigen	Anti-B, anti-D (if exposed)
A-positive	A antigen, D antigen	Anti-B
B-negative	[illegible]	ti-A, anti-D (if exposed)
B-positive	B antigen, D antigen	Anti-A
AB-negative	A and B antigens	Anti-D (if exposed)
AB-positive	A and B antigens, D antigen	No antibodies
O-negative	No antigens	Anti-A, anti-B, anti-D (if exposed)
O-positive	Rh antigen	Anti-A, anti-B

CLINICAL CORRELATION

A common example of this is an Rh-negative mother giving birth to an Rh-positive baby, which is only possible if the father was Rh-positive. During childbirth—and even sometimes during pregnancy—some fetal blood enters the maternal circulation. If the baby's blood is Rh-positive, and the mother's blood is Rh-negative, then the mother will make anti-Rh antibodies against the fetal red blood cells. At first, these anti-Rh antibodies are IgM in nature—so they're too big to cross the placenta. There is usually no harm to the fetus in the current pregnancy. However, mother's anti-Rh antibodies will eventually become IgG antibodies (which can cross the placenta). In any subsequent pregnancy with an Rh-positive fetus, mother's anti-Rh antibodies will cross the placenta and attack (hemolyze) the fetus' Rh-positive red blood cells. This is called hemolytic disease of the newborn (HDN), and it is marked by severe hemolytic anemia, jaundice, and hepatomegaly (enlarged liver) in the fetus. It can be fatal (Figure 2).

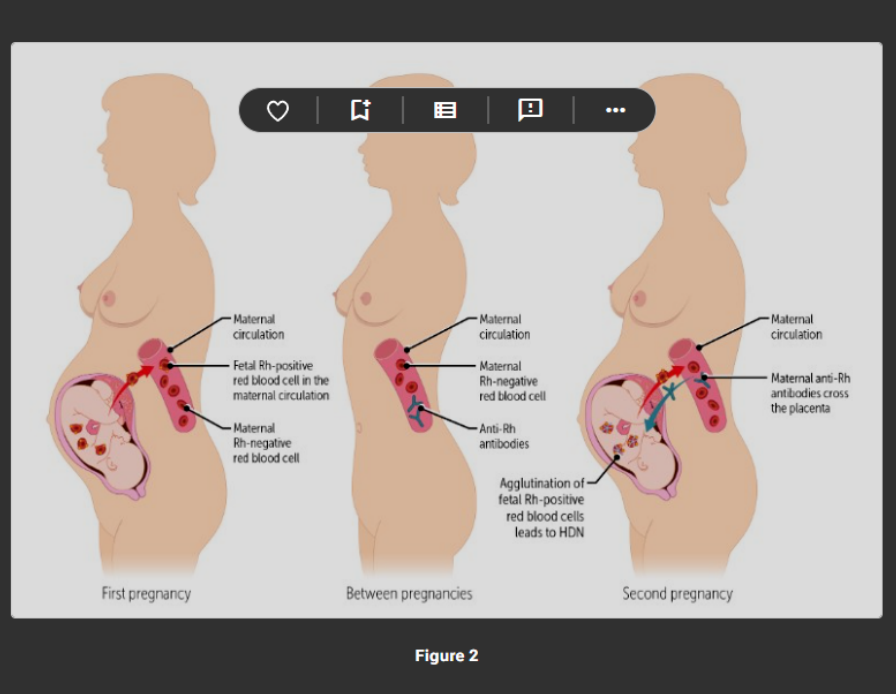
Maternal circulation
Fetal Rh-positive red blood cell in the maternal circulation
Maternal Rh-negative red blood cell
First pregnancy
Maternal circulation
Maternal Rh-negative red blood cell
Anti-Rh antibodies
Agglutination of fetal Rh-positive red blood cells leads to HDN
Between pregnancies
Maternal circulation
Maternal anti-Rh antibodies cross the placenta
Second pregnancy

Figure 2

CLINICAL CORRELATION

To prevent Rh-negative mothers from developing anti-D antibodies, we can give an injection called RhoGAM, which is a type of anti-D antibody that does not cross the placenta. If the fetus is Rh-positive, and some of the fetal RBCs make it into the mother's circulation, the RhoGAM anti-D antibodies will bind to the fetal RBCs, preventing the mother's immune system from "seeing" the Rh antigen. This effectively prevents formation of maternal anti-D antibodies, and it's given both during pregnancy and immediately after delivery, because fetal blood may enter maternal circulation at any time.

What Makes Antibodies Important for Transfusion?

Anti-A, anti-B, and anti-Rh (anti-D) antibodies are especially important when we talk about blood transfusions, a common procedure in which blood, or blood components from a donor, are transferred into a recipient's bloodstream. If the host has antibodies against the antigens on the transfused red blood cells, a severe—and even life-threatening—reaction can occur.

There are very specific ways in which blood types must be matched for a safe transfusion. The universal RBC donor has type O-negative blood; this is because there are no A, B, or D antigens on the surface of the RBC, so the blood will not prompt a retaliatory anti-A, anti-B, or anti-D response. On the flip side, an AB-positive patient has no antibodies against these antigens, so they can receive any type of blood—a "universal recipient."

Because patients whose plasma contains anti-A, anti-B, or anti-D antibodies will react with mismatched transfused RBCs, it is important to put the right blood into the right person! For example, if your patient has type A blood, they have anti-B antibodies and should only receive type A or type O blood. Rh-negative blood is given to Rh-negative patients, and Rh-positive or Rh-negative blood may be given to Rh-positive patients. Let's summarize (Table 3).

Table 3

Blood Type	Antibodies Present	Can Transfuse With
A-positive	Anti-B	A-positive, A-negative, O-positive, O-negative
A-negative	[illegible]	A-negative, O-negative
B-positive	Anti-A	B-positive, B-negative, O-positive, O-negative
B-negative	Anti-A, anti-D (if exposed)	B-negative, O-negative
AB-positive	None	All types
AB-negative	Anti-D (if exposed)	AB-negative, A-negative, B-negative, O-negative
O-positive	Anti-A, anti-B	O-positive, O-negative
O-negative	Anti-A, anti-B, anti-D (if exposed)	O-negative

What types of blood can be transfused into a patient with type AB-negative blood?

What Is a Blood Transfusion?

A blood transfusion is the process of transferring blood or blood components from one person, a donor, to another, the recipient. A patient might need a transfusion to replace blood lost during surgery or a traumatic injury (from a car crash); people with certain medical conditions, such as diseases that cause anemia, may also need transfusions. Although a blood transfusion is usually a safe procedure, many factors must be considered to ensure that the donor's blood and the recipient's blood are a match, to prevent adverse reactions.

Pre-Transfusion Testing

Blood donation and storage centers in the United States are regulated by the Food and Drug Administration to keep the blood supply safe. Before donation occurs, volunteers are asked to fill out questionnaires to assess their risk for certain transmittable diseases. After donation, blood products are tested for evidence of infections such as human immunodeficiency virus (HIV) and hepatitis B. Blood products that pass all safety tests are then ready for use by hospitals and other medical centers.

Prior to transfusion, more testing is performed to make sure that the donated blood is compatible with the patient. This pre-transfusion process consists of two main parts: (1) a type and screen and (2) a cross-match.

Type and Screen

The first step in pre-transfusion testing involves two tests: "typing" (which determines the patient's ABO/Rh blood group type) and "screening" (which determines whether the patient has serum antibodies against any other blood group antigens). Let's look at blood typing first.

Blood typing is a two-step process. First, the patient's red blood cells are tested for ABO and Rh antigens (this is called forward typing), and second, the patient's serum is tested for anti-ABO and anti-Rh antibodies (this is called reverse typing). Figure 3 illustrates this process in more detail.

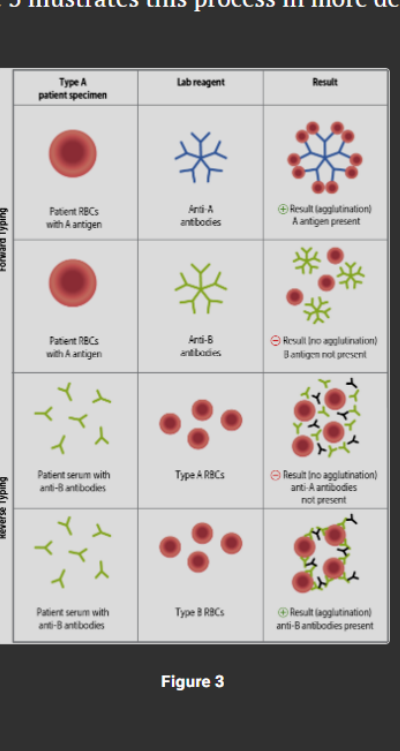
Type A patient specimen
Lab reagent
Result
Forward Typing
Reverse Typing
Patient RBCs with A antigen
Anti-A antibodies
⊕ Result (agglutination) A antigen present
Patient RBCs with A antigen
Anti-B antibodies
⊖ Result (no agglutination) B antigen not present
Patient serum with anti-B antibodies
Type A RBCs
⊖ Result (no agglutination) anti-A antibodies not present
Patient serum with anti-B antibodies
Type B RBCs
⊕ Result (agglutination) anti-B antibodies present

Figure 3

What would forward typing of a person with group A blood show?

After the blood typing procedure is complete, a blood screen is performed. In this assay, the patient's serum is tested against a small panel of RBCs with various non-ABO/Rh RBC antigens. These RBCs are purchased from a manufacturer and are tested to determine whether the patient's serum contains antibodies against various RBC antigens.

Cross-Matching

The second step in pre-transfusion testing is the cross-match, which is like a mini-transfusion: the patient's serum is combined with donor red blood cells (Figure 4). If agglutination or hemolysis occurs, that means the patient has an antibody against an antigen on the donor's RBCs, and the donor blood must not be transfused into the patient. This is the final step in testing for compatibility and is generally done immediately before the transfusion.

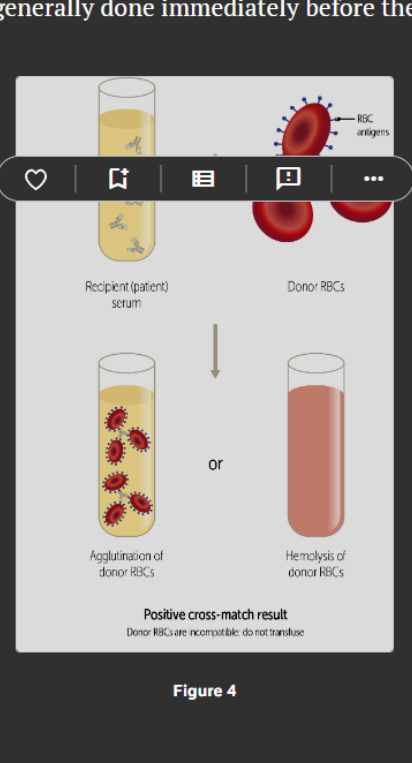
RBC antigens
Recipient (patient) serum
Donor RBCs
or
Agglutination of donor RBCs
Hemolysis of donor RBCs
Positive cross-match result
Donor RBCs are incompatible: do not transfuse

Figure 4

CLINICAL CORRELATION

Of note, the process of a crossmatch typically takes around 45 minutes but can be 1-2 hours or longer. In certain circumstances, patients cannot wait that long for blood. In those cases, the hospital can call on "**emergency blood release**", a protocol by which blood products are released from the blood bank immediately, bypassing standard testing, when a patient's life is at risk. Emergency blood release typically involves red blood cells that are O-negative (universal donor) and compatible with any patient.

However, as O-negative units are precious and can be in short supply, emergency release is typically only continued until the patient's blood type can be established and type-specific ur

What Are the Types of Blood Products (LO #4)?

According to the American Red Cross, every 2 seconds someone in the United States needs whole blood, RBCs, platelets, or plasma for surgery, cancer treatment, chronic illnesses, or traumatic injury. Whole blood is donated and then separated by centrifuge into red blood cells, platelets, and plasma.

The most commonly transfused blood products are whole blood, packed RBCs, platelets, granulocytes, cryoprecipitate, and fresh frozen plasma. These products and their indications are summarized in Table 4.

Table 4

Product	Indications	Notes
Packed RBCs	Severe anemia (blood hemoglobin <7 g/dL or <8 g/dL) with symptoms. Needed for increased oxygen-carrying capacity of blood.	1 unit of packed RBCs will raise the hemoglobin by approximately 1 g/dL.
Platelets	Medically significant bleeding related to thrombocytopenia (low platelet count). Different platelet count cut-offs are used depending on clinical indication (i.e. <50,000 for major surgery)	Platelets have ABO antigens on their surface, but they lack Rh antigens. Each unit of apheresis platelets should raise the platelet count by 30,000-50,000/mm³. Platelets are crucial to primary hemostasis.
Granulocytes	Uncontrolled infection (no [illegible] absolute neutrophil count <500/mm³.	
Fresh frozen plasma	Abnormal bleeding with coagulation factor deficiencies, warfarin over-anticoagulation, liver failure, thrombotic thrombocytopenic purpura.	Contains all coagulation factors.
Cryoprecipitate	Bleeding emergencies in patients with coagulation factor deficiencies of fibrinogen (fibrinogen <100).	Contains fibrinogen, factor VIII, VWF. Made by thawing fresh frozen plasma and collecting precipitate by centrifugation.
Whole blood	Large-volume traumatic hemorrhage.	

What is the main indication for transfusion of packed RBCs?

CLINICAL CORRELATION

"Massive Transfusion" refers to any situation where a patient is getting a lot of blood transfusions. Different definitions exist, with a common one being >10 units in 24 hours. **"Massive Transfusion Protocol" (MTP)** refers to rapid administration of large amounts of blood products (at least 6 units of PRBC) in fixed ratios (usually 1:1:1) for the management of hemorrhagic shock. Indications for activating a massive transfusion protocol include: setting of hemorrhage (traumatic, obstetric, etc.) and/or the pattern of blood product usage and response to treatment suggests that >6-10 units of blood products will be needed.

What Are Potential Complications of Blood Transfusion (LO #5)?

Patients often voice concern about blood transfusions because of the risk of infection with blood-borne pathogens and transfusion reactions. With advances in medical technology that allow for screening of blood after it is donated, infection from donor blood is now a very rare occurrence in developed countries. However, transfusions are not without risk. Let's look at some potential complications.

Infections

Transfusion recipients can become infected with pathogens contained in the donated blood. Although this is increasingly rare in developed countries, where, for example, HIV transmission from blood transfusion is now less than 1 in 1,000,000 patients, the screening may be less rigorous and the risk much higher in developing countries.

After whole blood is donated, it is screened for a multitude of infectious diseases. These include hepatitis B, hepatitis C, cytomegalovirus (CMV), HIV, and *Trypanosoma cruzi*. Blood that tests positive for any of these diseases is discarded.

CLINICAL CORRELATION

Cytomegalovirus (CMV) is a common viral pathogen that infects people of all ages. Healthy patients are able to fight off the initial CMV infection on their own; however, like the herpes or varicella viruses, CMV remains dormant in the body even after the symptoms of infection are gone. When an otherwise healthy donor gives blood that has latent CMV, the virus has the potential to reactivate and cause disease in the blood recipient. This can cause fatal cytomegalovirus in patients who are immunocompromised, such as transplant recipients who are taking multiple immunosuppressive drugs or premature infants. They are often not able to fight off the infection.

Because of this risk, many institutions now practice universal leukoreduction of their blood products. This removes the white blood cells that could potentially contain the CMV virus. These products, called "CMV-safe", are considered equivalent to products collected from CMV-negative donors.

When is it important to test donated blood for cytomegalovirus?

In addition to viruses, blood products may also be contaminated with bacteria. The source of the bacteria can be the donor's blood, the donor's skin, the phlebotomist's skin, or environmental contamination during production or packaging. Platelet transfusions have a higher risk of carrying bacterial pathogens because they are stored at room temperature. Bacterial contamination of blood products is extremely dangerous because it can result in a rapid progression to bacteremia and septic shock if not recognized and treated.

Transfusion Reactions

Despite careful pre-transfusion testing, reactions against transfused blood, known as transfusion reactions, still occur. It is important to monitor patients closely as they are undergoing transfusion for the earliest signs of any adverse reaction. Details of the most important transfusion reactions are summarized in Table 5.

Table 5

Reaction Type			Treatment
Acute hemolytic	Recipient antibodies against ABO antigens on donor RBCs. **DAT positive in the post-transfusion sample.**	Fever/chills (most common presenting symptom), back pain, hemoglobinuria, kidney failure, lung injury, disseminated intravascular coagulation (DIC), anemia, hypotension/shock, and death.	IV Hydration, Blood Pressure support, Diuresis
Delayed hemolytic	Recipient antibodies against minor (non-ABO) antigens on donor RBCs. Typically, an antibody that fades from detection over time, and patient is re-exposed to antigen on future transfusion	Mild anemia or fever, +/- jaundice; occurs 1 week or more after transfusion	Supportive care, steroids, identify underlying antibody
Allergic	Type 1 hypersensitivity reaction against allergens in donor blood.	Fever, urticaria, itching, wheezing. Anaphylactic shock in severe cases.	Antihistamines (may use for pre-medication), may require epinephrine/steroids for anaphylactic reaction
Febrile	Recipient antibodies against HLA antigens on donor leukocytes.	Fever, flushing.	Stop transfusion, may premedicate with anti-fever meds. Reduced incidence with leukoreduction
Transfusion-related acute lung injury	Mechanism uncertain; may involve donor antibodies attacking	Hypoxia, hypotension, respiratory failure 6	Supportive care; may require intubation
Transfusion-associated graft-versus-host disease	Donor T lymphocytes attack body tissues in an immunocompromised recipient.	Fever, rash, diarrhea hepatomegaly, death (80-90% fatal). Presents 7-10 days post-transfusion	Supportive care, immunosuppression; prevention with irradiation

Which blood group antigens are usually mismatched in acute hemolytic transfusion reactions?

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to HH, what is the likely diagnosis? How do you manage the condition?

Although HH's symptoms and physical exam findings could be secondary to her pneumonia, the timing of HH's clinical deterioration points toward acute hemolytic reaction secondary to blood transfusion. You recognize this as a medical emergency. You imm [illegible] er IV fluids and vasopressors. HH slowly improves, though her MICU stay is prolonged. You send the remaining blood to the blood bank, where it is determined that the blood for HH was mislabeled. In keeping with full disclosure and to prevent this from occurring again, you inform HH of the error and report the event to the quality and safety committee in the hospital.

Summary

- Blood type refers to the combination of blood group antigens present on a person's red blood cells, typically antigens from the ABO and Rh group.
- The ABO system includes three antigens: A, B, and H. The Rh system contains the D antigen, also known as the Rh antigen.
- The immune system can form antibodies against the antigens in both the ABO and Rh groups. Antibodies against the A and B antigens are naturally occurring, while antibodies against the Rh antigen are not.
- Blood typing is a two-step process in which the patient's red blood cells are tested for ABO and Rh antigens (forward typing), and second, the patient's serum is tested for anti-ABO and anti-Rh antibodies (reverse typing).
- Blood transfusion is the process of transferring blood or blood components from a donor to a recipient. It is important to ensure that the donor's blood and the recipient's blood are a match to prevent adverse reactions.
- The most commonly used blood products are packed red blood cells (used to treat severe, symptomatic anemia), platelets (used to treat bleeding caused by thrombocytopenia), granulocytes (used to treat uncontrolled infection in patients with neutropenia), fresh frozen plasma (used to treat bleeding with coagulation factor deficiencies), cryoprecipitate (used to treat bleeding wi [illegible] used to treat large volume traumatic hemorrhage).
- Potential complications of blood transfusion include infections and transfusion reactions, which can range from mild allergic reactions to severe hemolytic reactions.
- Pre-transfusion testing includes a type and screen to determine the patient's ABO/Rh blood group type and any serum antibodies against other blood group antigens, and a cross-match to ensure compatibility between the patient's serum and donor red blood cells.

Review Questions

1. Several minutes after starting a transfusion of packed red blood cells, a patient becomes hypotensive and short of breath and develops hives all over her body. Which of the following types of transfusion reaction is this patient likely experiencing?

A. ○ Acute hemolytic transfusion reaction
B. ● Allergic transfusion reaction
C. ○ Delayed hemolytic transfusion reaction
D. ○ Febrile nonhemolytic transfusion reaction
E. ○ Transfusion-related lung injury

Hide Explanation

The correct answer is an allergic (IgE-mediated) transfusion reaction (B). This patient is experiencing difficulty breathing or airway obstruction and hypotension from mast cell and histamine activation, causing vasodilation and urticaria—all of which are consistent with an anaphylactic reaction. Acute hemolytic transfusion reaction (A) is incorrect because these patients do not typically present with urticaria. Delayed hemolytic transfusion reaction (C) is incorrect because this patient began to experience symptoms minutes after her transfusion began. Delayed hemolytic reactions take about 3-10 days to occur. Febrile nonhemolytic transfusion reaction (D) is incorrect because this patient did not have a fever. Febrile reactions often present with fever, chills, headache, and flushing. Transfusion-related lung injury (E) is a severe form of acute respiratory distress syndrome that occurs with blood transfusion; although this patient was having difficulty breathing, the clinical picture of dyspnea plus hives plus hypotension should point you toward an IgE-mediated reaction.

2. Which type of transfusion reaction is most common?

A. ○ Acute hemolytic
B. ● Allergic
C. ○ Delayed hemolytic
D. ○ Transfusion-associated graft-versus-host disease
E. ○ Transfusion-related acute lung injury

Hide Explanation

The correct answer is allergic (B). These are IgE-mediated type 1 hypersensitivity responses that lead to itching, hives, fever, and wheezing. Anaphylaxis occasionally occurs. Along with nonhemolytic febrile reactions, they are the most common of the transfusion reactions. Acute (A) and delayed (C) hemolytic reactions are much less common and lead to anemia and fever, but acute reactions are much more severe. Transfusion-associated graft-versus-host disease (D) is rare and occurs when donor lymphocytes attack host tissue, especially in immunocompromised hosts who cannot ward off this attack. Transfusion-related acute lung injury (E) is very uncommon and presents with respiratory distress.

3. A 45-year-old man develops recurrent rectal bleeding from a colonic diverticulum. His hematocrit drops, and he becomes tachycardic and dizzy. His physicians prepare for an urgent blood transfusion. Which of the following lab tests requires the use of actual donated red blood cells?

A. ○ ABO testing
B. ○ Antibody screening
C. ● Cross-matching
D. ○ D antigen testing
E. ○ Rh testing

Hide Explanation

The correct answer is cross-matching (C). After determining the patient's ABO (A) and Rh (D) blood type, the patient's serum is mixed with the actual RBCs to be used in the donation to see if they cross-react. This cross-matching is the only step that uses the donated RBCs. The remaining options (A, B, D, E) are all tests done using recipient blood and lab reagents, not donor RBCs.

4. Patients with which blood type may have both anti-A and anti-Rh antibodies?

A. ○ A-negative
B. ○ A-positive
C. ● B-negative
D. ○ B-positive
E. ○ O-positive

Hide Explanation

The answer is B-negative blood (C). B-negative red cells have the only the B antigen (not the A or Rh antigens) on their surface. Antibodies against ABO antigens are naturally occurring—so a patient with type B blood will automatically make antibodies against th [illegible] urally occurring—so to make these antibodies, a person must be Rh-negative and must be exposed to Rh-positive blood. An A-negative patient (A) may have anti-Rh antibodies (if there was prior exposure to Rh blood) but would not have anti-A antibodies. An A-positive patient (B) would have only anti-B antibodies (not anti-A or anti-Rh antibodies). A B-positive patient (D) would have anti-A antibodies, but not anti-Rh antibodies. An O-positive patient (E) would have anti-A and anti-B antibodies (but not anti-Rh antibodies).

5. Which of the following would be the best blood product to use in a bleeding patient with low fibrinogen?

A. ○ Fresh frozen plasma (FFP)
B. ● Cryoprecipitate
C. ○ Granulocytes
D. ○ Packed RBCs
E. ○ Platelets

Hide Explanation

The correct answer is cryoprecipitate (B). FFP (A) does contain fibrinogen, but it would be diluted in a large volume of plasma. Granulocytes (C) are used in patients who have an uncontrolled infection and low absolute neutrophil count. Packed RBCs (D) are used in patien [illegible] Hgb <7 g/dL). Platelets (E) are used in patients with medically significant bleeding related to thrombocytopenia; it would not help correct low fibrinogen.

How would you rate this Brick?

References

- Fung, Mark K, Anne Eder, Steven L. Spitalnik, and Connie M. Westhoff. *Technical Manual.*, 2017. Print.
- McPherson, Richard A, and Matthew R. Pincus. *Henry's Clinical Diagnosis and Management by Laboratory Methods.*, 2017. Print.
- Hoffman, R, Benz EJ, Silberstein L, et al. *Hematology: Basic Principles and Practice*. 8th ed. Elsevier; 2023.

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ Recognize the antigens of the ABO and Rh blood groups.
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Objective Confidence: 5/5